

	Document number:	Document name:	Date:
	SGR.D.GFS.0059.v1.0 E	Cloudgate Mini Black Box Connection	03.03.2026
	Function owner:	Function verifictor:	Function approver:
	KPO- Fleet & Operations Manager	Group Service Manager	Group Fleet & Rental Operations Director

Black Box connection information Suivo Pulse

Crane number	Brand /Type	Date	Supplier	MHE/ Vans/ Pool cars
		Technician	SUIVO / GFS	
Serial number crane	IMEI number Black Box		Maintenance order (SAP)	

Pin	Signal	Type	Description
1	GND	GND	Ground for VCC and IO
2	VCC	VCC	+6 to +31VDC charge input and power supply
3	RS-232 Tx	0	RS-232 Tx
4	Analog In 5	I	Analog (Vehicle Contact) Input #5 (0..35V)
A	RS-232-Rx	I	RS-232 Rx
B	CAN-H	I/O	CAN-H
C	CAN-L	I/O	CAN-L
D	Analog in/out 2	I/O	Configurable Analog input or output #2 (0..35V)



Connection for data collection:

* Read before starting installation! General
- Only connect Engine Canbus on MHE, and if available.

rules for Engine running data collection -----
MHE prior 2011 will be - generally - connected using standard 24 v (inputs) signals (As done previously with Multicap)

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- Read the installation manual before starting the installation!!
- All Yellow coloured cells are to be filled!!



Connection points:								
	Pin	Naming	Data type	Info page schemas	Wire color	Crane connector name (From schemas)	Signal checks	Signal function on Platform
	2	24V	KL 30 - Battery supply				Supply from battery KL 30+. Before master switch! Write voltage value:	Power supply for the tracker when the machine is not in used
	1	Ground	KL 31 - Chassis ground				Ground / masa connection	Ground
	4	Ignition signal	KL 15 - Ignition signal - Digital In 5				Ignition signal KL+15. Note that BB needs to have ignition signal for data collection. Write voltage value:	Wake up data collection

CAN 1 or I/O		Engine data (Only for MHE if available!)	CanBus - Connected to the engine can			Can H :	Installation tool - Sartel- check & note Baudrate:	Engine running status Hour counter function Engine data values collection if CanBus connected Diagnostic engine if CanBus connected
	B	CAN H	☐			Can L :	Installation tool - Sartel- check & note Configuration:	
	C	CAN_L						
	D	Digital In 1	I/O standard D+ ☐				Installation check with multimeter - Write Voltage value:	

Information working hours
Working hours Primary Engine / Lower Engine :
Specified position of BB in the crane:
Take pictures of the following: IMEI sticker: ☐ Location of BB in the crane: ☐ Connection of Can 2 (if connected): ☐ Engine id plate: ☐

Remarks
- Responsible fleet manager must send picture and correctly filled sheet to bb.sartel@sarens.com with cc to Suresh.